



ISO 9001:2015

ISO 13485

PRECISION PLASTICS ENGINEERING

Corporate Capabilities & Services Catalogue

A fully integrated single-roof partner for tooling, high-precision injection moulding, extrusion, metrology, assembly, and cleanroom medical device manufacturing.

● Product Design & DFM

● High-Precision Tool Room

● High-Volume Injection Moulding

● ISO Class 8 Cleanrooms



CORPORATE PROFILE & CATALOGUE INDEX

Vidhata Plastics India Pvt. Ltd. is a fully integrated plastics manufacturing company. With over 20 years of experience, we provide comprehensive end-to-end solutions under a single roof, helping our customers reduce lead times, elevate quality, and simplify supply chain operations.



Corporate Focus

- **Mission:** To deliver world-class plastic manufacturing solutions through continuous innovation, uncompromising quality standards, and customer-focused execution.
- **Vision:** To become a globally trusted manufacturing partner for medical, pharmaceutical, electronics, and industrial sectors.

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01. ENGINEERING & TOOLROOM CAPABILITIES

Our state-of-the-art in-house Tool Room transforms ideas into highly precise moulds. By integrating engineering, design, and manufacturing under one roof, we eliminate intermediate supply chain delays and enforce strict tolerances.



● Design & DFM

- **CAD/CAM Modeling:** 3D product design, reverse engineering, and prototyping using industry-leading platforms.
- **DFM & Mold Flow Analysis:** Early optimization of gate locations, wall thickness, cooling, and sink mark prevention.

● Tool Manufacturing

- **Mould Types:** Multi-cavity injection moulds, extrusion dies, polyurethane (PU) casting moulds, prototype tools.
- **Precision Machining:** Ultra-fine tolerances down to ± 5 microns for critical medical and electronic parts.

EQUIPMENT CATEGORY	MAKE & SPECIFICATIONS	APPLICATION / PURPOSE
CNC Milling	DMG Mori, Makino CNC Machining Centers	High-speed, high-precision mould cavity cutting
Wire Cut EDM	Fanuc Wire EDM System	Intricate geometry and precise core profiling
Die Sinking EDM	Charmilles EDM Machines	Deep cavity sparks and high-finish details
Precision Grinding	High-accuracy Surface and Cylindrical Grinding	Flatness, parting line alignment, and fitting



02. ADVANCED INJECTION MOULDING & EXTRUSION

We operate high-volume, automated production lines capable of processing a wide range of engineering plastics. Our machinery is configured for high repeatability, minimized waste, and fast cycle times.



● Injection Moulding

- **Capacity:** Range of Haitian injection moulding machines from 120T to 250T clamping force.
- **Blow Moulding:** 60T dedicated Injection Blow Moulding (IBM) line for containers and closures.

● Extrusion & PU processing

- **Extrusion Lines:** Polycarbonate (PC) and medical grade tubing extrusion with precise dimensional control.
- **Polyurethane Systems:** DESMA processing systems and BASF Elastogran systems.

PROCESS CAPABILITY	MATERIAL SPECIALIZATION	KEY FEATURES
High-Volume Injection	PC, PMMA, ABS, PP, Nylon, POM, PBT, PS	Automated material dryers, dehumidifiers, pick-and-place robots
Injection Blow Moulding	HDPE, LDPE, PP, PET	Ideal for pharmaceutical bottles and leak-proof closures
Profile Extrusion	Polycarbonate, PVC, Medical Grade Polyurethane	LED diffusers, batten housings, medical tubing
Polyurethane Casting	Soft and Hard Polyurethane Elastomers	BASF chemistry, high wear-resistant industrial/shoe parts



03. ISO 13485 CLEANROOM MEDICAL MANUFACTURING

We operate ISO Class 8 (100,000) cleanroom facilities strictly monitored for dust, temperature, and microbiological contamination. Our processes conform to medical device manufacturing standards.



● Cleanroom Moulding

- **Sterile Injection Moulding:** Low-particulate, contamination-free moulding of critical fluid pathways.
- **Medical Materials:** Certified processing of USP Class VI, medical-grade polycarbonate, acrylic, and PP.

● Medical Product Segments

- **Dialysis Components:** Dialysis housing elements, blood ports, and end-caps.
- **Fluid Management:** I.V. cannula components, catheter manifolds, drip chambers, luer adapters, and closures.

PRODUCT CATEGORY	MATERIALS USED	STANDARDS & APPROVALS
IVD & Diagnostics	High-flow Polystyrene, PMMA, Medical Polycarbonate	CE-ready, ISO 13485 compliant, DNA/RNase-free
Drug Delivery Devices	Cyclic Olefin Copolymer (COC), Polycarbonate, ABS	Device Master Records (DMR) maintained, QMSR compliant
Dialysis Parts	Medical Polycarbonate, Polypropylene, Elastomers	Biocompatible materials, particulate-controlled lines
closures & closures	HDPE, Child Resistant Closures, Flip-Off Caps	Cleanroom blow moulding, strict leak/autoclave testing



04. ASSEMBLY, PACKAGING & SECONDARY SERVICES

Vidhata is more than a moulder. We offer extensive post-moulding capabilities including multi-part mechanical assembly, leak testing, medical-grade packaging, and supply chain management.



● Assembly Services

- **Mechanical & Manual Assembly:** Cleanroom lines for complex disposable devices and electronic housings.
- **Bonding & Welding:** Ultrasonic welding, solvent bonding, and gluing for airtight, high-strength joints.

● Packaging & Logistics

- **Sterile Packaging:** Automated pouch sealing, Tyvek packing, and blister packaging inside cleanrooms.
- **Sterilization Support:** Coordination of EO (Ethylene Oxide) or Gamma sterilization protocols.

SERVICE OFFERING	CAPABILITIES & METHODS	QUALITY CONTROL
Leak & Flow Testing	Ateq pressure-decay testers, multi-channel flow checking	100% inline inspection
Ultrasonic Welding	High-frequency plastic welding of polycarbonate/ABS	Joint integrity shear test
Blister & Tyvek	Medical-grade heat sealing, sterile barrier creation	Peel-strength testing
Documentation	Device Master Records (DMR), Design History Files (DHF)	100% traceability



05. QUALITY ASSURANCE, METROLOGY & PORTFOLIO

Quality is built into every stage of our project lifecycle. Our state-of-the-art metrology lab and strict quality systems ensure that every batch of parts meets the exact engineering drawing specifications.



● Metrology & Lab Assets

- **Dimensional Inspection:** Trimos height gauges, Mitutoyo profile projectors, microscopes.
- **Material & Product Verification:** Moisture analyzers, melt flow indexers (MFI), stability chambers, ovens.

● Industry Success Cases

- **LED Lighting:** Over 30 Million units manufactured annually for Philips and Eveready.
- **Electronics:** Over 12 Million high-finish mobile charger housings delivered to Reliance Jio.
- **Defense:** 24 high-complexity EVM moulds designed and built for ECIL.

"Vidhata matched Chinese pricing models on tool manufacturing while significantly beating them on quality, lead-times, and on-the-ground technical support." — Leading Indian Electronics Client

VIDHATA PLASTICS INDIA PVT. LTD.

Precision Plastics, From Design to Delivery

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